

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

WORKED SOLUTIONS

CLASS

INDEX NUMBER

CHEMISTRY

9729/01

Paper 1 Multiple Choice

27 June 2017

1 hour

Additional Materials: *Data Booklet*
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

1	D	6	B	11	B	16	B	21	D	26	D
2	C	7	C	12	D	17	B	22	B	27	D
3	B	8	C	13	D	18	C	23	C	28	A
4	B	9	A	14	C	19	D	24	B	29	C
5	C	10	D	15	B	20	C	25	D	30	C

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 A given mass of ideal gas occupies a volume V and exerts a pressure p at 30°C .

At which temperature will the same mass of the ideal gas occupy a volume $\frac{V}{3}$ and exert a pressure $2p$?

A 20°C

C 202°C

B 20 K

D 202 K

Answer: **D**

$$pV = nRT$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

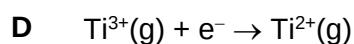
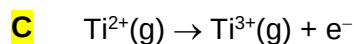
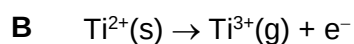
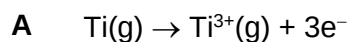
$$\frac{P_1 V_1}{T_1} = \frac{2P_1 \frac{V_1}{3}}{T_2}$$

$$T_2 = \frac{2}{3} T_1$$

$$\text{Original } T = 30 + 273 = 303\text{ K}$$

$$\text{New } T = \frac{2}{3} \times 303 = 202\text{ K}$$

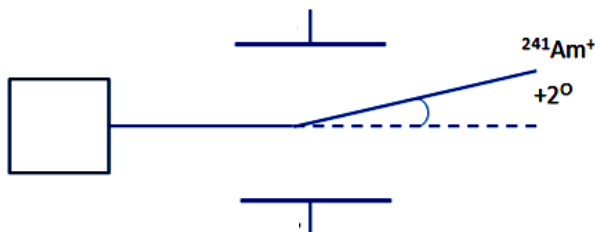
- 2 Which equation corresponds to the third ionisation energy of titanium (Ti)?



Answer: **C**

Third ionization energy of Ti is defined as the amount of energy required to remove 1 mole of electrons from one mole of doubly charged gaseous Ti (Ti^{2+}) to form one mole of triply charged gaseous Ti (Ti^{3+}).

- 3 A sample of the element Americium (Am) was vaporised, ionised and passed through an electric field. It was observed that a beam of $^{241}\text{Am}^+$ particles gave an angle of deflection of $+2^\circ$.



Assuming an identical set of experimental conditions, by what angle would a beam of $^{32}\text{S}^-$ particles be deflected?

- A $+15.1^\circ$
B -15.1°
 C $+30.1^\circ$
 D -30.1°

Answer: B

angle of deflection $\propto \frac{\text{charge}}{\text{mass}}$

angle of deflection of $^{241}\text{Am}^+ = +2^\circ$

$$\text{For } ^{241}\text{Am}^+, \frac{q}{m} = \frac{1}{241}$$

$$k \frac{1}{241} = (2)$$

$$k = 482$$

$$\text{For } ^{32}\text{S}^-, \frac{q}{m} = \frac{1}{32}$$

$$\text{angle of deflection of } ^{32}\text{S}^- = \frac{1}{32} \times 482 = -15.1^\circ$$

- 4 Which of the following statements describes a phenomenon which **cannot** be explained by hydrogen bonding?
- A Ice floats on water.
B The boiling point of carboxylic acid increases with increasing relative molecular mass.
 C 2-nitrophenol is more volatile than 4-nitrophenol.
 D Ethanoic acid molecules form dimers when dissolved in benzene.

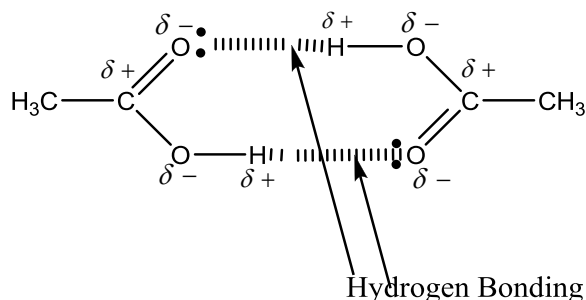
Answer: **B**

Option **A** is incorrect as in ice, each H_2O molecule forms the maximum of **four** hydrogen bonds with 4 other H_2O molecules. This results in an **open structure** of ice and hence ice is **less dense** than water and so, floats on water.

Option **B** is correct as the boiling point of carboxylic acid increases with increasing M_r due to **stronger instantaneous dipole-induced dipole interactions between the non-polar R group** as the size of electron cloud of R group increases and hence become more polarisable.

Option **C** is incorrect as in 2-nitrophenol, due to close proximity between $-\text{OH}$ and $-\text{NO}_2$ groups, they can be used for formation of **intramolecular hydrogen bonds**. Hence, there are **fewer available sites** for **intermolecular hydrogen bonds**. Thus, 2-nitrophenol has a lower boiling point (and hence more volatile) as **lesser energy is required** to overcome the **less extensive hydrogen bonds** between molecules.

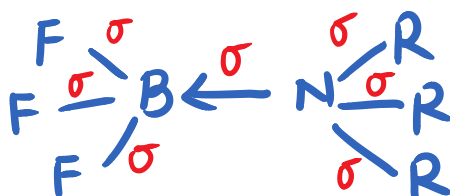
Option **D** is incorrect as in benzene, CH_3COOH molecules can form a dimer via hydrogen bonding.



- 5 A tertiary amine, R_3N , reacts with boron trifluoride, BF_3 to give an addition product. Which of the following statements is **not** true?
- A R_3N acts as a Lewis base.
 - B The product is a polar molecule.
 - C** There are six σ bonds in the product.
 - D The product contains a dative covalent bond.

Answer: **C**

Addition product:



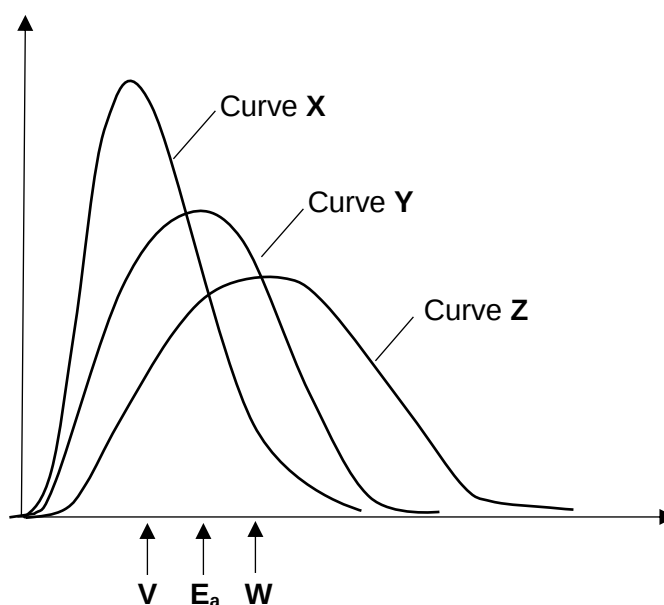
Option **A** is incorrect as R_3N is a Lewis base as it **donates an electron pair** to BF_3 .

Option **B** is incorrect as the molecule is polar due to presence of **polar bonds** ($\text{C}-\text{N}$, $\text{B}-\text{F}$ and $\text{B}-\text{N}$) and the **dipole moments do not cancel out**.

Option **C** is correct as there is a **minimum of 7 σ bonds** in the addition product (not taking into account the $\text{C}-\text{H}$ and $\text{C}-\text{C}$ σ bonds present in the R groups).

Option **D** is correct as there is a **dative covalent bond from N to B**.

- 6 The curve Y and the value E_a represent the distribution of energies of the molecules and the activation energy for an uncatalysed gaseous reaction.



What is a possible outcome if the reaction is catalysed?

- A The distribution of energies will be given by curve X and the activation energy by value V
- B** The distribution of energies will be given by curve Y and the activation energy by value V.
- C The distribution of energies will be given by curve Y and the activation energy by value W.
- D The distribution of energies will be given by curve Z and the activation energy by value W.

Answer: **B**

When catalyst is added, the graph distribution of energies of molecules will not change, it is affected only by temperature. Hence the curve should be Y.
However, when catalyst is added E_a will become lowered, hence value V.

- 7 If the rate of decay of a radioactive isotope decrease from 300 counts per minute to 37.5 counts per minute after 1 day, what is its half-life?
- A 4 hours
 - C** 8 hours
 - B 6 hours
 - D 12 hours

Answer: **C**

First half-life occurs when 300 counts per minute is reduced to 150 counts per minute.
Second half-life occurs when 150 counts per minute is reduced to 75 counts per minute.
Third half-life occurs when 75 counts per minute is reduced to 37.5 counts per minute.

3 half-life in 24 hours means each half-life is 8 hours.

8 Which of the following statements does **not** describe a reaction at equilibrium?

- A Forward and backward reactions occur at equal rate.
- B The system must be closed.
- C** K_c increases as the reaction progresses.
- D Concentrations of reactants and products are constant.

Answer: **C**

At dynamic equilibrium, rate of forward reaction = rate of backward reaction, hence there is no change in concentration of reactants and products.

K_c is only affected by **temperature changes**.

9 Use of the *Data Booklet* is relevant to this question.

The enthalpy change of formation of carbon monoxide and carbon dioxide are given below.

$$\Delta H_f \text{ CO} = -110 \text{ kJmol}^{-1}$$

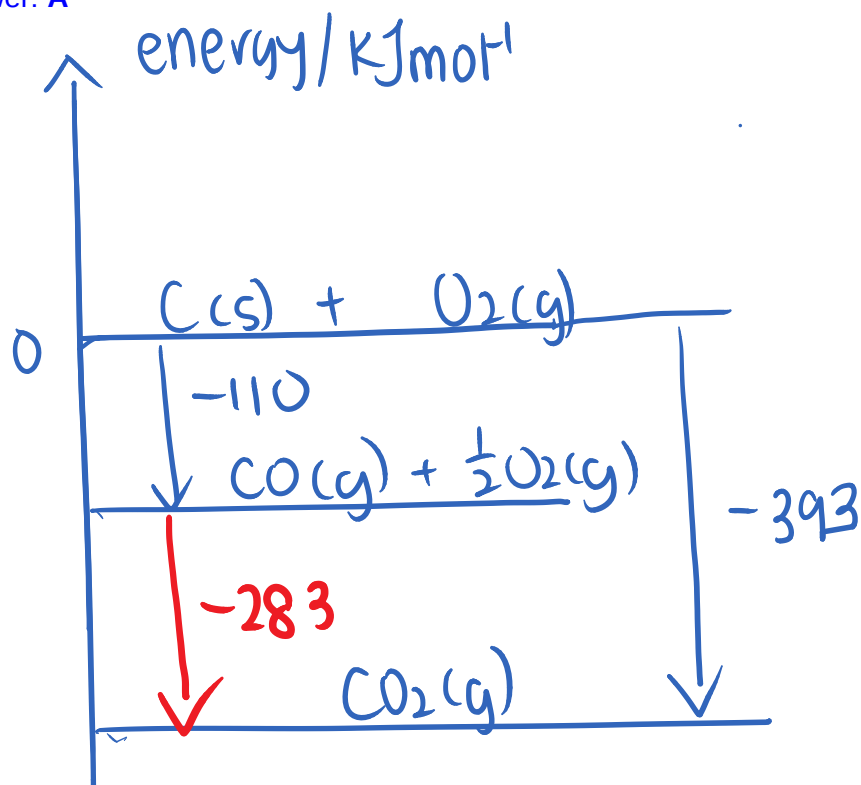
$$\Delta H_f \text{ CO}_2 = -393 \text{ kJmol}^{-1}$$

Which of these statements are correct?

- 1 $\text{CO}_2(\text{g})$ has lower energy content than $\text{CO}(\text{g})$.
- 2 The enthalpy change of combustion of carbon is -393 kJmol^{-1} .
- 3 $\text{CO}_2(\text{g})$ is formed exothermically from $\text{CO}(\text{g})$.
- 4 A larger amount of energy is required to atomise $\text{CO}_2(\text{g})$ than $\text{CO}(\text{g})$.

- A** 1, 2, 3 and 4
- B 1, 2 and 3 only
- C 1, 2 and 4 only
- D 2 and 3 only

Answer: A



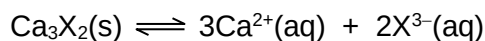
Option 1 is correct as from the energy level diagram, $\text{CO}_2\text{(g)}$ has a lower energy content than CO(g) .

Option 2 is correct as $\Delta H_c(\text{C}) = \Delta H_f(\text{CO}_2) = -393 \text{ kJ mol}^{-1}$ from the diagram

Option 3 is correct as from the energy level diagram, $\text{CO}_2\text{(g)}$ is formed exothermically from CO(g) . $[-283 \text{ kJ mol}^{-1}]$

Option 4 is correct as atomisation means to produce free gaseous C and O atoms (an endothermic process). More energy is required to atomise $\text{CO}_2\text{(g)}$ than CO(g) as $\text{CO}_2\text{(g)}$ has a lower energy content.

- 10 A sparingly soluble calcium salt ionises in aqueous solution according to the equation given:



If the solubility product K_{sp} of Ca_3X_2 is S , what is the value of the concentration of $\text{Ca}^{2+}(\text{aq})$ at equilibrium?

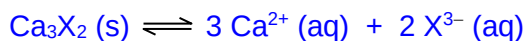
A $S^{\frac{1}{2}}$

B $\left[\frac{S}{108} \right]^{\frac{1}{5}}$

C $\left[\frac{3S}{5} \right]^{\frac{1}{2}}$

D $\left[\frac{9S}{4} \right]^{\frac{1}{5}}$

Answer: **D**



$$K_{\text{sp}} = [\text{Ca}^{2+}]^3 [\text{X}^{3-}]^2$$

$$S = (3x)^3 (2x)^2$$

$$= (27x^3) (4x^2)$$

$$= 108 x^5$$

$$x = \left(\frac{S}{108} \right)^{\frac{1}{5}}$$

$$[\text{Ca}^{2+}] \text{ at eqm} = 3x = 3 \left(\frac{S}{108} \right)^{\frac{1}{5}} = \left(\frac{243S}{108} \right)^{\frac{1}{5}} = \left[\frac{9S}{4} \right]^{\frac{1}{5}}$$

- 11 The K_{sp} of AgCl and AgI are $1.80 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$ and $8.3 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$ respectively. Which of the following statements is correct when equal volumes of $1 \times 10^{-4} \text{ mol dm}^{-3}$ aqueous AgNO_3 was added to a mixture containing $3.0 \times 10^{-6} \text{ mol dm}^{-3} \text{ BaCl}_2$ and $3.0 \times 10^{-6} \text{ mol dm}^{-3} \text{ BaI}_2$?

- A** AgCl is precipitated only.
B AgI is precipitated only.
C AgCl is precipitated followed by AgI .
D AgI is precipitated followed by AgCl .

Answer: **B**

$$[\text{Ag}^+] = [\text{AgNO}_3] = 1 \times 10^{-4} \text{ mol dm}^{-3}$$

$$[\text{Cl}^-] = 2 \times [\text{BaCl}_2] = 6 \times 10^{-6} \text{ mol dm}^{-3}$$

$$[\text{I}^-] = 2 \times [\text{BaI}_2] = 6 \times 10^{-6} \text{ mol dm}^{-3}$$

Take note that the mixture consists of equal volumes of AgNO_3 and BaCl_2 . Hence, the new concentration will be halved.

$$[\text{Ag}^+] \text{ in mixture} = 0.5 \times 10^{-4}$$

$$[\text{Cl}^-] \text{ in mixture} = 3 \times 10^{-6}$$

$$\text{IP of AgCl} = (0.5 \times 10^{-4})(3.0 \times 10^{-6}) = 1.5 \times 10^{-10}$$

$\text{IP} < K_{\text{sp}} \rightarrow \text{AgCl}$ will not ppt out

$$\text{IP of AgI} = (0.5 \times 10^{-4})(3.0 \times 10^{-6}) = 1.5 \times 10^{-10}$$

$\text{IP} > K_{\text{sp}} \rightarrow \text{AgI}$ will ppt out

- 12 The dissociation constant, K_w , for the ionisation of water, $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$, at different temperatures is given below.

Temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
0	1.15×10^{-15}
25	1.00×10^{-14}
50	5.50×10^{-14}

What can be deduced from this information?

- A Only at 25 °C are $[\text{H}^+]$ and $[\text{OH}^-]$ equal.
 B The equilibrium lies furthest to the right at 0 °C.
 C The forward reaction is exothermic.
 D The pH of pure water decreases with temperature.

Answer: **D**

Option A is incorrect as the $[\text{H}^+]$ and $[\text{OH}^-]$ of water are equal at all temperatures.

Option B is incorrect as the value of K_w is the smallest at 0 °C. Hence, equilibrium lies most to the left.

Option C is incorrect as K_w increases with temperature, the forward reaction is favoured.

Option D is correct as an increase in temperature favours the endothermic reaction, hence the forward reaction is endothermic.

Option D:

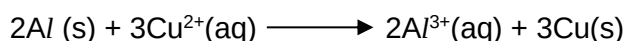
$$K_w = [\text{H}^+][\text{OH}^-]$$

$$K_w = [\text{H}^+]^2$$

$$\text{pH} = -\log \sqrt{K_w}$$

as K_w increases with temperature, pH will decrease.

- 13 Calculate the standard Gibbs free energy change, ΔG° for the following reaction:



- A -386 kJ mol^{-1}
 B -579 kJ mol^{-1}
 C $-1045 \text{ kJ mol}^{-1}$
 D $-1158 \text{ kJ mol}^{-1}$

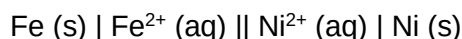
Answer: **D**

$$E_{\text{cell}} = +0.34 - (-1.66) = +2.00 \text{ V}$$

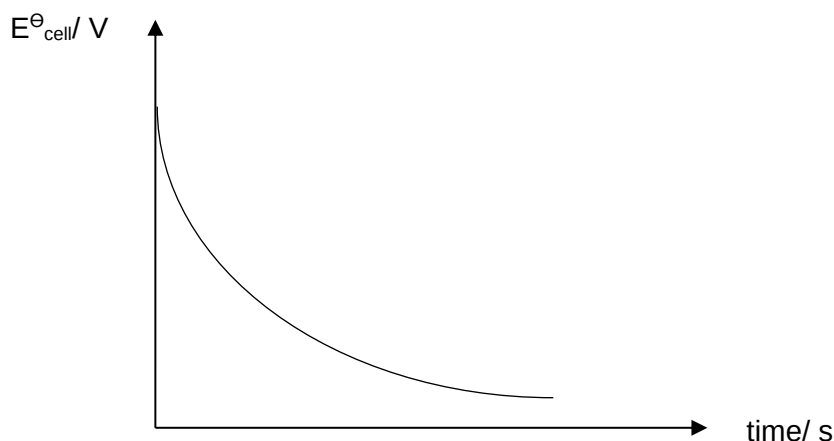
$$\Delta G^\circ = -nFE_{\text{cell}} = -6 \times 96500 \times 2.00 = -1158000 \text{ J mol}^{-1}$$

$$= \underline{\underline{-1158 \text{ kJ mol}^{-1}}}$$

- 14 An experiment is carried out with the following cell.



The following graph of cell potential against time was obtained when a change was continuously made to the half-cell.



What continuous change could produce these results?

- A Add nickel (II) chloride to the nickel half-cell.
- B Add aqueous cyanide ions to the iron half-cell.
- C** Add water to the nickel half-cell.
- D Increases the surface area of iron immersed in the solution.

Answer: **C**

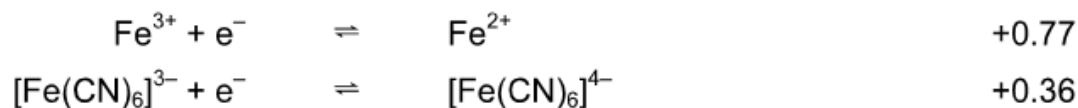
$$E_{\text{cell}} = E_{\text{Ni}^{2+}/\text{Ni}} - E_{\text{Fe}^{3+}/\text{Fe}^{2+}} \text{ (graph states that } E_{\text{cell}} \text{ should decrease)}$$

Option **A** is incorrect as



When nickel (II) chloride is added to the nickel half-cell, $[\text{Ni}^{2+}]$ increase, eqm position shift right. $E_{\text{Ni}^{2+}/\text{Ni}}$ is more positive (incorrect)

Option **B** is incorrect as

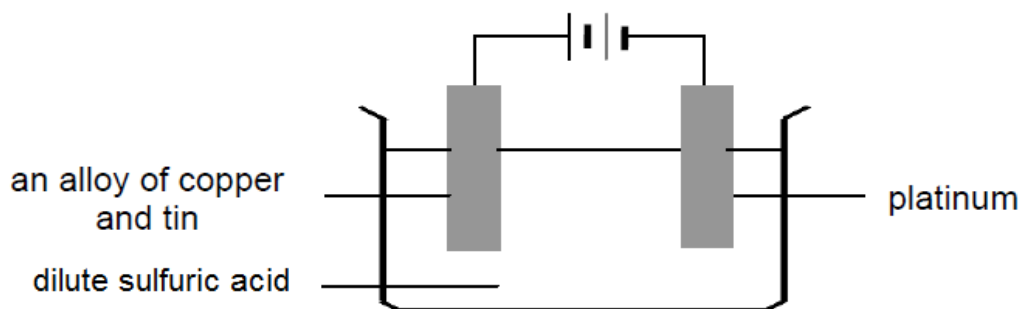


When aqueous cyanide ions is added to the iron half-cell. $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}$ is less positive (incorrect)

Option **C** is correct as when water is added to the nickel half-cell, $[\text{Ni}^{2+}]$ decrease, eqm position shift left. $E_{\text{Ni}^{2+}/\text{Ni}}$ is less positive (correct)

Option **D** is incorrect as increases in the surface area of iron affects rate not E values.

- 15 The circuit shown in the diagram was set up.



Which reactions will occur at the electrodes?

<i>anode reaction</i>	<i>cathode reaction</i>
A Oxygen gas is evolved.	Hydrogen gas is evolved.
B Tin dissolves preferentially.	Hydrogen gas is evolved.
C Copper dissolves preferentially.	Copper is deposited.
D Copper and tin both dissolve.	Sulfur dioxide gas is evolved.

Answer: **B**

At the anode: (oxidation will occur)

Species present: Cu and Sn

Consider

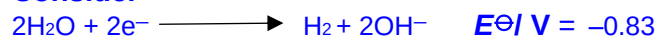


Based on the $E^{\ominus}/V$ values, a less positive value indicates that tin will be selectively oxidised.

At the cathode: (reduction will occur)

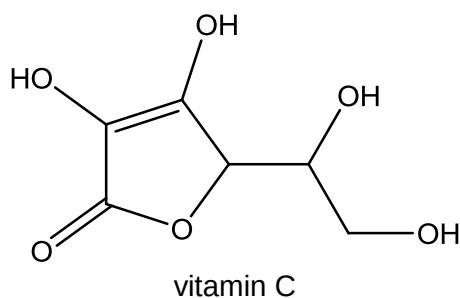
Species present: H_2O and H^{+}

Consider



Based on the $E^{\ominus}/V$ values, a more $E^{\ominus}/V$ positive value indicates that H^{+} will be selectively reduced to produce H_2 gas.

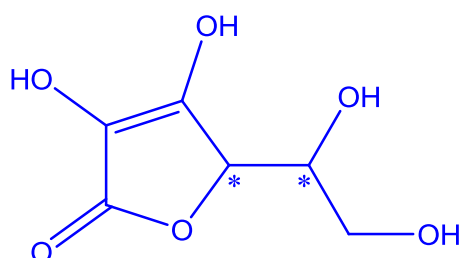
- 16 The diagram shows the structure of vitamin C.



How many stereoisomers are there in one molecule of vitamin C?

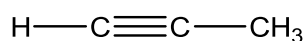
- A** 2 **B** 4 **C** 8 **D** 16

Answer: **B**



2 chiral centres so number of enantiomers are $2^2 = 4$

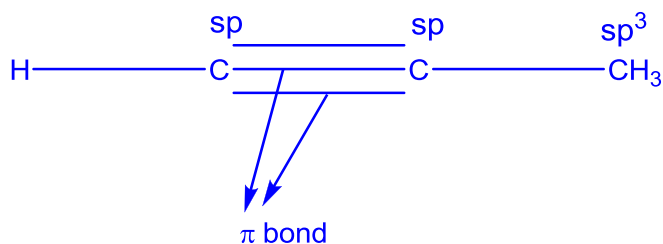
- 17 Propyne, C_3H_4 , has the following structure.



Which row correctly describes the bonding and hybridisation in a molecule of propyne?

	number of π bonds	number of sp C atoms	number of sp^2 C atoms
A	1	1	1
B	2	2	0
C	2	2	1
D	3	3	0

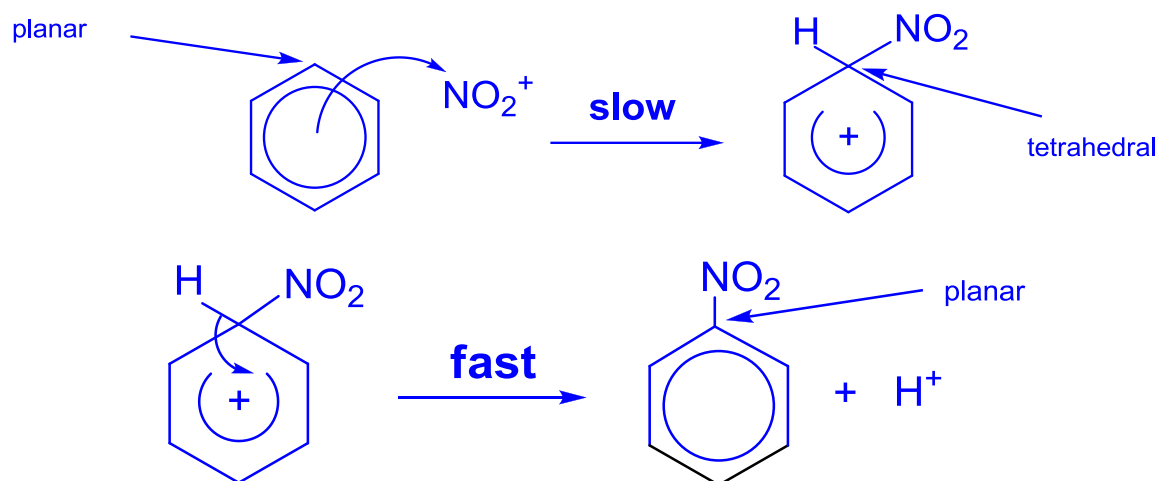
Answer: **B**



- 18 During the nitration of benzene, a nitro group substitutes at a carbon atom. Which one of the following gives the arrangement of the bonds at this carbon atom during the reaction?

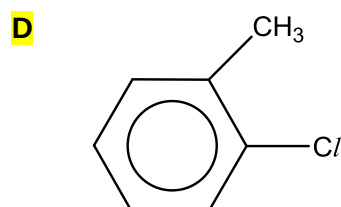
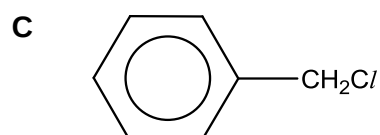
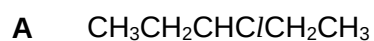
	<i>at the start of the reaction</i>	<i>in the intermediate complex</i>	<i>at the end of the reaction</i>
A	planar	planar	planar
B	planar	tetrahedral	tetrahedral
C	planar	tetrahedral	planar
D	tetrahedral	planar	tetrahedral

Answer: **C**



- 19 When a halogen compound **S** was boiled under reflux for some time with silver nitrate in a mixture of ethanol and water, little or no precipitate was seen.

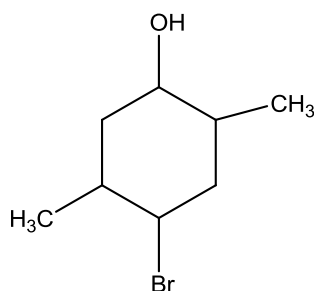
Which of the following formulae could represent **S**?



Answer: **D**

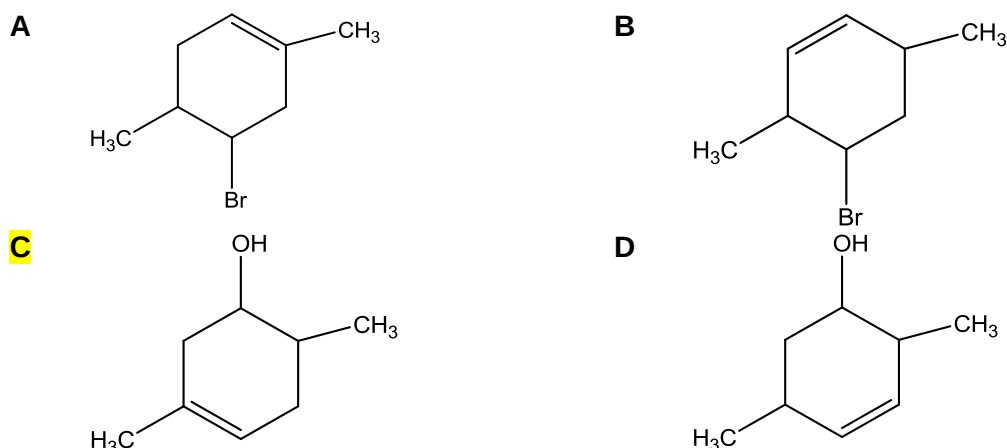
Given that there is little or no ppt seen, it shows that the C-Cl bond is strong, hence it is referring to the Cl bonded to a benzene ring as the p orbital of Cl will overlap with the π orbital of the benzene ring.

- 20 Compound **P** was heated with ethanolic potassium hydroxide.



compound **P**

Which of the following would be the major product?



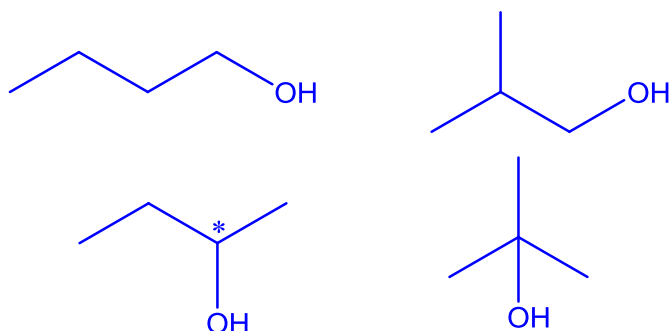
Answer: **C**

Ethanol KOH is the reagent to eliminate H and Br → **A** and **B** will be wrong
Using Saytzeff rule whereby the "poor get poorer"

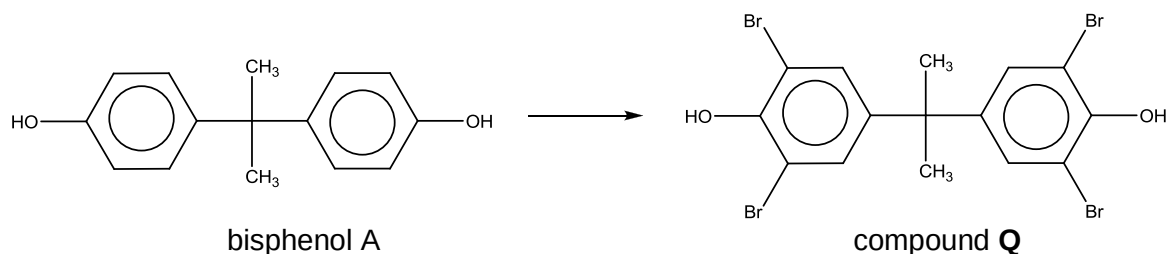
- 21 How many isomers (including both structural isomers and stereoisomers) with molecular formula $C_4H_{10}O$ liberates hydrogen on reaction with sodium?

A 2 **B** 3 **C** 4 **D** 5

Answer: **D**



- 22 Bisphenol A was used to make products such as plastic polycarbonate baby bottles and food containers. It is now regarded as toxic and has been withdrawn from use.



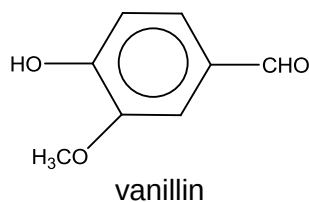
Which reagent will convert bisphenol A into compound **Q**?

- A** $AlBr_3(s)$
B $Br_2(aq)$
C $HBr(g)$
D $NaBr(aq)$

Answer: **B**

Based on the product, the reaction is electrophilic substitution. Since the starting reactant is a phenol, catalyst will not be necessary. Hence $\text{Br}_2(\text{aq})$ will be sufficient.

- 23 Vanillin is the active ingredient of vanilla.



Which of the following will be observed with vanillin?

- 1 Warm acidified potassium dichromate (VI) turns green.
- 2 2,4-dinitrophenylhydrazine reagent gives an orange precipitate.
- 3 A yellow precipitate is formed on warming with aqueous alkaline iodine.

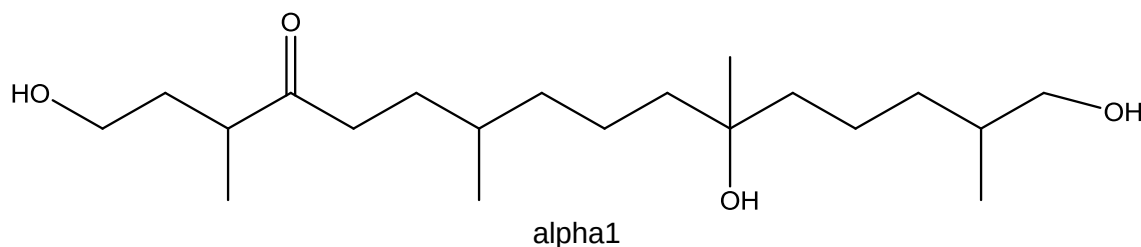
- A 1 only
 B 2 only
C 1 and 2 only
 D 2 and 3 only

Answer: **C**

Option 1 and 2 are correct as Vanillin has the aldehyde group which can be oxidised by potassium dichromate and react with 2,4-DNPH.

Option 3 is not correct as there is no presence of $\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$ group in vanillin hence it does not react with aqueous alkaline iodine.

- 24 The mould *Phytophthora* damages many plants, destroying agricultural crops such as potatoes. A hormone-like compound called alpha1 regulates the reproduction of all species of *Phytophthora*. The structure of alpha1 is now known, giving scientist a key to the possible future eradication of the mould.



Which of the following reagents will react with alpha1?

- 1 Br_2
 - 2 SOCl_2
 - 3 H_2/Pt
- A 1 and 2 only

- B** 2 and 3 only
C 1 and 3 only
D 1, 2 and 3 only

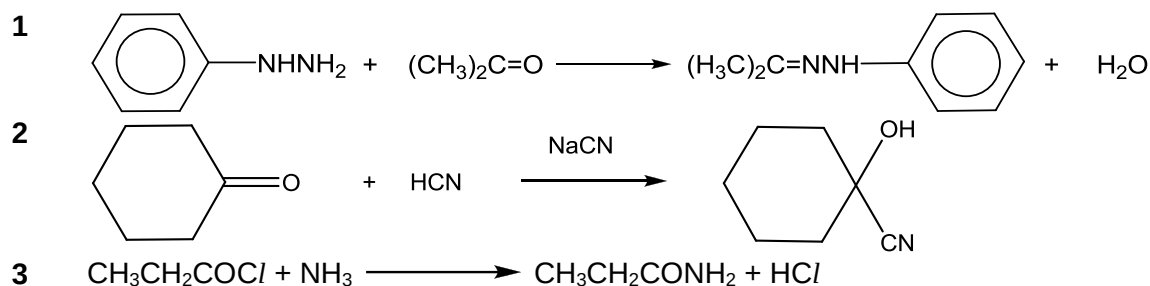
Answer: **B**

Alpha1 is not able to react with aqueous Br_2 since there is no $\text{C}=\text{C}$.

OH group on Alpha1 is able to react with SOCl_2 to form halogenoalkane.

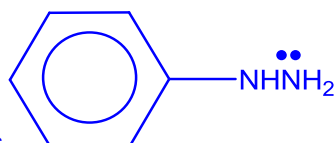
Carbonyl group on alpha1 can be reduced by H_2/Pt to form secondary alcohol.

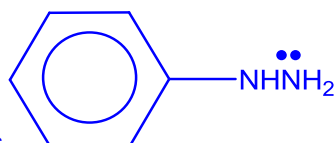
25 Which transformations involve a nucleophile?



- A** 1 only
B 2 and 3 only
C 1 and 2 only
D 1, 2 and 3 only

Answer: **D**



Option 1 is correct as  is the nucleophile. Similar to 2,4-DNPH
 Option 2 is correct as CN^- is the nucleophile. Reaction is nucleophilic addition.
 Option 3 is correct as NH_3 is the nucleophile. Reaction is nucleophilic acyl substitution.

26 Which of the following properties are identical for the two enantiomers of 2-hydroxypropanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$?

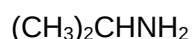
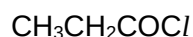
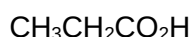
- 1 ΔH_f°
 2 $\text{p}K_a$
 3 melting point

- A** 2 only
B 3 only
C 1 and 2 only
D 1, 2 and 3 only

Answer: **D**

Enantiomers will only differ in their biological properties. Given that they have the same functional group, their chemical and physical properties should remain the same.

- 27 When organic compounds **E**, **F**, **G** and **H** are added separately to water, solutions of increasing pH values are obtained. The possible identities of compounds **E** to **H** (not necessarily in that order) are given below.



Which is the correct set of identities of compounds **E**, **F**, **G** and **H**?

	E	F	G	H
A	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CH}_2\text{COC}l$	$\text{CH}_3\text{CH}_2\text{NH}_2$	$(\text{CH}_3)_2\text{CHNH}_2$
B	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CH}_2\text{COC}l$	$(\text{CH}_3)_2\text{CHNH}_2$	$\text{CH}_3\text{CH}_2\text{NH}_2$
C	$\text{CH}_3\text{CH}_2\text{COC}l$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	$(\text{CH}_3)_2\text{CHNH}_2$	$\text{CH}_3\text{CH}_2\text{NH}_2$
D	$\text{CH}_3\text{CH}_2\text{COC}l$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CH}_2\text{NH}_2$	$(\text{CH}_3)_2\text{CHNH}_2$

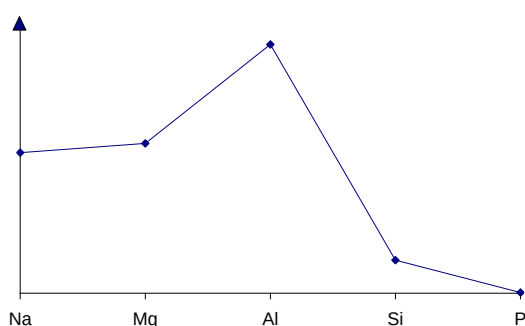
Answer: **D**

Between $\text{CH}_3\text{CH}_2\text{COC}l$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ is a weaker acid as one mole of $\text{CH}_3\text{CH}_2\text{COC}l$ will form one mole of $\text{CH}_3\text{CH}_2\text{COOH}$ and one mole of HCl in aqueous solution while $\text{CH}_3\text{CH}_2\text{COOH}$ will only partially dissociate to form H^+ . $\text{CH}_3\text{CH}_2\text{COC}l$ will have a lower pH than $\text{CH}_3\text{CH}_2\text{COOH}$.

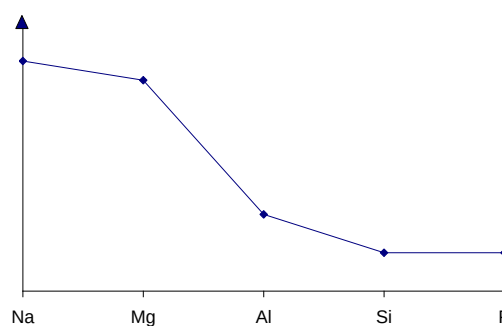
$(\text{CH}_3)_2\text{CHNH}_2$ is a stronger base as compared to $\text{CH}_3\text{CH}_2\text{NH}_2$ as it has one more methyl electron donating group, hence the lone pair is more available to accept H^+ . $(\text{CH}_3)_2\text{CHNH}_2$ will have a higher pH as compared to $\text{CH}_3\text{CH}_2\text{NH}_2$.

- 28 The graphs below show the variation in two properties of the elements Na to P and their compounds.

Graph I



Graph II



Which properties are illustrated in Graphs I and II?

	Graph I	Graph II
A	electrical conductivity of the element	pH of the chloride when added to water
B	electrical conductivity of the element	pH of the oxide when added to water
C	melting point of the element	pH of the chloride when added to water

	melting point of the element	pH of the oxide when added to water
D		

Answer: **A**

Graph 1:

Electrical conductivity increases from Na to Al due to the increasing amount of delocalised electrons. Si is a semiconductor / metalloid hence it is able to conduct electricity. Phosphorus is a non-metal with a simple covalent structure. Hence it is unable to conduct electricity.

Melting point of graph should increase from Na to Al and peak at Si as Si has a giant covalent structure. Hence, options C and D are eliminated.

Graph 2:

NaCl - pH 7

MgCl₂ - pH 6.5

AlCl₃ - pH 3

SiCl₄ - pH 2

PCl₅ - pH 2

Na₂O - pH 13

MgO - pH 10

Al₂O₃ - not soluble hence pH 7

SiO₂ - not soluble hence pH 7

P₄O₁₀ - pH 2

- 29** Which of the following elements is expected to show the greatest tendency to form some covalent compounds?

- A** Barium
- B** Calcium
- C** Magnesium
- D** Potassium

Answer: **C**

The metal cation with the highest charge density will have the greatest polarising power and hence the greatest tendency to form covalent compounds.

Charge density = charge / ionic radius

Since Mg has a charge of +2 and the smallest size, it has the highest charge density

- 30** Why is hydrogen iodide a stronger acid than hydrogen chloride?

- A** A molecule of hydrogen chloride is more polar than a molecule of hydrogen iodide.
- B** The enthalpy change of formation of hydrogen iodide is greater than that of hydrogen chloride.
- C** The covalent bond in the hydrogen iodide molecule is weaker than that in the hydrogen chloride molecule.

- D** The dissociation of hydrogen chloride molecules is suppressed by the stronger permanent dipole–permanent dipole interactions.

Answer: **C**

For HI and HCl to form acids, the H – X bond must be broken and the molecules need to dissociate in water to form H^+ and X^- ions.

Hence, the molecule with the weaker H – X bond will be able to dissociate to a greater extent to form more H^+ ions to be the stronger acid.